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CAPSTONE 1 — Password and Multi-Factor Authentication Hygiene Kit

Project Question

How can AI help design a practical, easy-to-follow password and MFA habit-building programme for students?

What you have to do

1. Choose the target audience (first-year college students or hostel residents).
2. Research password attacks (credential stuffing, brute force, reuse) and MFA methods using at least two AI research tools.
3. Collect and verify reliable sources (NIST SP 800-63B, CERT-In, CISA, official password-manager security documentation).
4. Create a research summary and a password and MFA habit checklist.
5. Write 6 short scenarios (e.g. reused password, shared OTP) with the correct safe response for each.
6. Write an 800-word beginner guide on passphrases, password managers and MFA types.
7. Create a poster or infographic comparing SMS OTP, authenticator app and security key.
8. Create a 5-slide talk and a 10-question quiz with answer key.
9. Automate one step (e.g. a weekly security-habit reminder drafted from the checklist).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: never collect or test real passwords, use only dummy accounts, avoid advice that weakens security.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + verification + image generation + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Leonardo.ai, Gamma, Make, Notion; plus Python (entropy calculator script on dummy data) and Google Sheets.

Final submission

- Research report
- Password and MFA habit checklist
- Scenario set
- Beginner guide
- Infographic
- Quiz and answer key
- Presentation
- Python entropy demo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 2 — Deepfake and AI Misinformation Awareness Pack

Project Question

How can AI be used responsibly to teach people to spot deepfakes and AI-generated misinformation?

What you have to do

1. Choose the target audience (school students, college students or general public).
2. Research deepfake types, detection cues and real-world incident patterns using at least two AI research tools.
3. Collect and verify reliable sources (PIB Fact Check, CERT-In, MeitY advisories, peer-reviewed papers via Consensus and Elicit).
4. Create a research summary and a spot-the-fake checklist.
5. Write 6 fictional "fake news" cards (clearly labelled as training examples, no real people).
6. Write a 1,000-word explainer article on how deepfakes work and how to verify media.
7. Generate 5 sample AI images with Leonardo.ai/Bing and annotate the visible AI artefacts.
8. Create a 6-slide presentation and a 10-question quiz with answer key.
9. Automate one step (e.g. a weekly fact-check tip drafted from the checklist).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: never create deepfakes of real people, label all synthetic media, discuss consent and misuse.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + image generation + verification + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, Consensus, NotebookLM, Leonardo.ai, Bing Image Creator, Gamma, Zapier/Make.

Final submission

- Research report
- Spot-the-fake checklist
- Fake-news cards
- Explainer article
- Annotated AI image set
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 3 — UPI and Digital Payment Safety Guide for Senior Citizens

Project Question

How can AI help create simple, large-print safety material that protects senior citizens from digital payment scams?

What you have to do

1. Choose the target audience (senior citizens and their family caregivers).
2. Research common UPI, OTP, KYC and fake-call scam patterns using at least two AI research tools.
3. Collect and verify reliable sources (RBI, NPCI, CERT-In and cybercrime.gov.in public advisories).
4. Create a research summary and a "stop and check" safety checklist.
5. Write 6 fictional scam call/message scripts labelled "training samples" with the correct response.
6. Write a 600-word large-print guide in plain language (also produce a Hindi or regional-language version).
7. Create a poster or one-page infographic with big icons and a helpline reminder.
8. Create a 5-slide family workshop and an 8-question quiz with answer key.
9. Automate one step (e.g. a weekly WhatsApp-ready safety tip draft).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: no real phone numbers, banks or victims, check translations with a native speaker, no financial advice.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + translation + image generation + quiz design + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Leonardo.ai, Gamma, ElevenLabs (audio tips), Zapier/Make.

Final submission

- Research report
- "stop and check" safety checklist
- Scam scripts
- Large-print guide
- Poster
- Quiz and answer key
- Audio tips
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 4 — Secure Coding Guide for Beginners (OWASP Top 10)

Project Question

How can AI help beginners understand common web vulnerabilities and write safer code, without producing attack tools?

What you have to do

1. Choose the target audience (second-year CSE students learning web development).
2. Research OWASP Top 10 categories and safe coding practices using at least two AI research tools.
3. Collect and verify reliable sources (OWASP, NIST SSDF, MDN Web Docs, official framework security guides).
4. Create a research summary and a secure-coding checklist.
5. Write 5 insecure-versus-secure code snippet pairs (e.g. SQL injection prevention, input validation) for a local practice app.
6. Write a 1,000-word guide explaining each risk with a defensive fix.
7. Create a one-page cheat sheet infographic of the OWASP Top 10 in simple words.
8. Create a 6-slide talk and a 10-question quiz with answer key.
9. Automate one step (e.g. a GitHub Action or script that checks code against the checklist).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: no working exploits, practise only on your own local lab or OWASP Juice Shop/DVWA, verify AI-written code before use.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + code explanation + verification + documentation + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, DeepSeek, Gamma, GitHub; plus VS Code, Python and OWASP Juice Shop in a local VM.

Final submission

- Research report
- Secure-coding checklist
- Code snippet pairs
- Guide
- Cheat sheet
- Quiz and answer key
- Presentation
- GitHub repository
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 5 — Incident Response Playbook for a Small Organisation

Project Question

How can AI help draft and stress-test a simple incident response playbook for a small office or college lab?

What you have to do

1. Choose the target audience (a small business, college lab or student club).
2. Research incident response phases (prepare, detect, contain, recover, review) using at least two AI research tools.
3. Collect and verify reliable sources (NIST SP 800-61, CERT-In reporting guidelines, CISA small-business resources).
4. Create a research summary and an incident response checklist and contact tree template.
5. Create a one-page playbook for 3 incidents (lost laptop, malware alert, account takeover).
6. Write a 3-scenario tabletop exercise with facilitator notes (all fictional).
7. Draft incident report and communication templates (internal notice, post-incident review).
8. Create a 6-slide briefing and a 10-question quiz with answer key.
9. Automate one step (e.g. a Zapier/Make flow that turns an incident form into a draft report).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: fictional organisation only, no real breach data, AI must not replace professional or legal guidance.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + process design + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Gamma, Zapier/Make, Notion, GitHub.

Final submission

- Research report
- Incident response checklist and contact tree template
- Playbook
- Tabletop exercise
- Report templates
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 6 — DPDP Act and Data Privacy Awareness for Students

Project Question

How can AI help explain data privacy rights and responsibilities in simple language for students?

What you have to do

1. Choose the target audience (college students and student-club volunteers).
2. Research India's Digital Personal Data Protection Act, GDPR basics and everyday privacy risks using at least two AI research tools.
3. Collect and verify reliable sources (MeitY and official gazette text, EU GDPR official site, CERT-In).
4. Create a research summary and a personal data privacy checklist.
5. Write a plain-language 1,000-word explainer comparing DPDP Act and GDPR basics.
6. Create 6 everyday scenarios (app permissions, consent forms) with the privacy-safe choice.
7. Create a poster or infographic of "My data rights".
8. Create a 6-slide presentation and a 10-question quiz with answer key.
9. Automate one step (e.g. weekly privacy tip drafts from the checklist).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: AI is not legal advice, no real personal data in prompts, flag outdated law summaries.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + legal-text summarisation + verification + image generation + quiz design + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, Consensus, NotebookLM, Leonardo.ai, Gamma, Make.

Final submission

- Research report
- Personal data privacy checklist
- Explainer
- Scenario set
- Infographic
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 7 — Social Media Privacy and Digital Footprint Toolkit

Project Question

How can AI help students understand and reduce their digital footprint on social platforms?

What you have to do

1. Choose the target audience (college students and early-career job seekers).
2. Research digital footprint, oversharing risks, location leaks and profile-based social engineering using at least two AI research tools.
3. Collect and verify reliable sources (CERT-In, CISA, platform privacy-help pages, consumer-protection advisories).
4. Create a research summary and a privacy settings audit checklist.
5. Create a fictional persona and show how oversharing could be misused (training example only).
6. Write an 800-word guide with step-by-step privacy settings review.
7. Create 5 social-media posts and an infographic on "Think before you post".
8. Create a 5-slide presentation and a 10-question quiz with answer key.
9. Automate one step (e.g. a monthly privacy-review reminder drafted from the checklist).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: never profile real people or run OSINT on real individuals, use fictional personas only.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + image generation + verification + quiz design + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Leonardo.ai, Gamma, Zapier/Make, Notion.

Final submission

- Research report
- Privacy settings audit checklist
- Persona case study
- Guide
- Posts and infographic
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 8 — Ransomware Readiness Guide for Small Businesses

Project Question

How can AI help prepare a small business to prevent, survive and recover from ransomware?

What you have to do

1. Choose the target audience (a small shop, clinic or startup with non-technical staff).
2. Research ransomware attack stages, backup strategies (3-2-1) and recovery planning using at least two AI research tools.
3. Collect and verify reliable sources (CISA #StopRansomware, CERT-In advisories, NIST, vendor threat reports).
4. Create a research summary and a ransomware readiness checklist.
5. Write a 1,000-word non-technical article on how ransomware works and prevention basics.
6. Draw a diagram of a safe backup plan (3-2-1) as an infographic.
7. Draft a one-page recovery plan template and an employee do/don't sheet.
8. Create a 6-slide presentation and a 10-question quiz with answer key.
9. Automate one step (e.g. a backup-test reminder workflow).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: no malware code or samples, no ransom-payment advice presented as certain, use fictional business.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + diagram design + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Leonardo.ai, Gamma, Zapier/Make, Google Sheets.

Final submission

- Research report
- Ransomware readiness checklist
- Article
- Backup infographic
- Recovery template
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 9 — Safe Public Wi-Fi and Basic Network Security Guide

Project Question

How can AI help explain network risks and safe habits using only sample packet captures and a lab setup?

What you have to do

1. Choose the target audience (college students who use campus, cafe and hostel Wi-Fi).
2. Research public Wi-Fi risks, HTTPS, VPN basics and common network attacks (conceptual) using at least two AI research tools.
3. Collect and verify reliable sources (CISA, CERT-In, Wireshark official docs, IETF/Mozilla HTTPS guides).
4. Create a research summary and a safe-Wi-Fi checklist.
5. Analyse a provided sample capture in Wireshark and have AI explain the packets (verify the explanation yourself).
6. Write a 900-word guide on HTTP vs HTTPS, DNS and VPN in simple words.
7. Create an infographic: "5 things to do before joining public Wi-Fi".
8. Create a 6-slide presentation and a 10-question quiz with answer key.
9. Automate one step (e.g. a Python script that summarises a capture's protocol counts).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: capture only your own lab traffic or official sample files, never sniff others' networks.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + technical explanation + verification + image generation + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Gamma, Leonardo.ai, GitHub; plus Wireshark (sample captures), Python and a local VM.

Final submission

- Research report
- Safe-Wi-Fi checklist
- Packet analysis notes
- Guide
- Infographic
- Quiz and answer key
- Presentation
- Python script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 10 — Python Learning Pathway for First-Year Students

Project Question

How can AI act as a patient tutor to build a structured 4-week Python learning pathway?

What you have to do

1. Choose the target audience (first-year CSE students with no programming background).
2. Research effective beginner learning paths and common Python misconceptions using at least two AI research tools.
3. Collect and verify reliable sources (official Python docs, Python tutorial, CS education papers via Consensus/Elicit).
4. Create a research summary and a weekly learning-outcomes checklist.
5. Create a 4-week plan with daily 30-minute tasks.
6. Write 20 graded practice problems with solutions (verify all by running the code).
7. Write a 1,000-word "common mistakes" article with fixed examples.
8. Create a 6-slide intro deck and a 15-question quiz with answer key.
9. Automate one step (e.g. a daily practice-problem email draft generated from the plan).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: avoid copying solutions blindly, run and test all AI code, credit sources.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + learning design + code explanation + verification + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, Make, GitHub; plus Python and Google Colab.

Final submission

- Research report
- Weekly learning-outcomes checklist
- 4-week plan
- Practice problem set
- Mistakes article
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 11 — Visual Study Guide for Data Structures and Algorithms

Project Question

How can AI help turn DSA topics into clear visual explanations, practice sets and revision notes?

What you have to do

1. Choose the target audience (second-year CSE students preparing for exams and placements).
2. Research arrays, linked lists, stacks, queues, trees, graphs, sorting and Big-O using at least two AI research tools.
3. Collect and verify reliable sources (textbook references, official language docs, university lecture notes).
4. Create a research summary and a DSA revision checklist with time complexities.
5. Write plain-language explanations with analogies for 8 core topics.
6. Generate and verify code in Python or C++ for each topic, with test cases.
7. Create complexity cheat-sheet infographics and diagrams.
8. Create a 7-slide revision deck and a 15-question quiz with answer key.
9. Automate one step (e.g. a daily "one DSA concept" note draft).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: verify complexity claims, do not submit AI code as original work without understanding it.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + technical explanation + code generation + verification + image generation + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, Leonardo.ai, GitHub, Notion; plus Python/C++ and Google Colab.

Final submission

- Research report
- DSA revision checklist with time complexities
- Concept notes
- Tested code repo
- Cheat sheets
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 12 — DBMS and SQL Practice Companion

Project Question

How can AI help students learn database design and SQL through guided practice and error explanation?

What you have to do

1. Choose the target audience (third-semester CSE students).
2. Research ER modelling, normalisation, joins, indexing and transactions using at least two AI research tools.
3. Collect and verify reliable sources (official PostgreSQL/MySQL/SQLite docs, university lecture notes).
4. Create a research summary and a normalisation and SQL query checklist.
5. Design a sample database (e.g. library or clinic, fictional data) with an ER diagram.
6. Write 25 SQL practice queries from easy to advanced with expected outputs (run them).
7. Create a normalisation explainer with a worked example from 1NF to 3NF.
8. Create a 6-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a script that turns SQL error messages into explanation drafts).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: fictional data only, verify AI-written queries, no real patient or student records.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + technical explanation + code verification + diagram design + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub; plus SQLite/PostgreSQL and Google Colab.

Final submission

- Research report
- Normalisation and SQL query checklist
- ER diagram and schema
- Query set
- Normalisation explainer
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 13 — Operating Systems Concepts Explainer Pack

Project Question

How can AI help simplify OS concepts such as scheduling, memory and deadlocks using visuals and simulations?

What you have to do

1. Choose the target audience (third-semester CSE students).
2. Research process scheduling, memory management, paging and deadlocks using at least two AI research tools.
3. Collect and verify reliable sources (official Linux documentation, standard OS textbooks, university notes).
4. Create a research summary and an OS concept revision checklist.
5. Explain 6 OS topics using real-life analogies.
6. Write Python simulations for FCFS, SJF and Round Robin and compare results (verify manually).
7. Create diagrams or infographics for paging and deadlock conditions.
8. Create a 6-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a script that generates scheduling tables for a given set of processes).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: verify simulation results against hand calculations, credit sources.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + technical explanation + code verification + image generation + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, Leonardo.ai, GitHub; plus Python and a Linux VM.

Final submission

- Research report
- OS concept revision checklist
- Concept notes
- Simulation code
- Diagrams
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 14 — Computer Networks Revision Kit (OSI and TCP/IP)

Project Question

How can AI help students master networking layers, addressing and protocols through examples and labs?

What you have to do

1. Choose the target audience (third-year CSE students).
2. Research OSI and TCP/IP layers, IP addressing and subnetting, DNS, HTTP and TCP using at least two AI research tools.
3. Collect and verify reliable sources (RFC summaries, Cisco/Mozilla free learning pages, Wireshark docs).
4. Create a research summary and a layer-by-layer protocol checklist.
5. Write layer-wise explanations with an everyday analogy (courier or post office).
6. Create 15 subnetting problems with step-by-step solutions (verify with a subnet calculator).
7. Analyse a sample Wireshark capture to match packets to layers.
8. Create a 7-slide deck and a 15-question quiz with answer key.
9. Automate one step (e.g. a Python subnet-practice generator).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: capture only lab or sample traffic, verify all numeric answers.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + technical explanation + calculation verification + image generation + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, Leonardo.ai, GitHub; plus Wireshark (sample captures) and Python.

Final submission

- Research report
- Layer-by-layer protocol checklist
- Layer notes
- Subnetting set
- Packet analysis
- Quiz and answer key
- Presentation
- Python generator
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 15 — Machine Learning for Beginners Mini-Course

Project Question

How can AI help create a beginner-friendly, math-light introduction to machine learning with working examples?

What you have to do

1. Choose the target audience (second or third-year CSE students new to ML).
2. Research supervised vs unsupervised learning, overfitting and model evaluation using at least two AI research tools.
3. Collect and verify reliable sources (scikit-learn docs, Google ML Crash Course, papers via Consensus/Elicit).
4. Create a research summary and an ML concepts and metrics checklist.
5. Write 6 short lessons with analogies and no heavy maths.
6. Build 2 small notebooks (e.g. iris classification, house price regression) on public datasets.
7. Create a visual guide to confusion matrix, precision and recall.
8. Create a 7-slide deck and a 15-question quiz with answer key.
9. Automate one step (e.g. a script that creates a model-report summary from metrics).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: use only open datasets, report limitations honestly, avoid overstating accuracy.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + technical explanation + code verification + image generation + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, Hugging Face, GitHub; plus Python, Google Colab and Kaggle datasets.

Final submission

- Research report
- ML concepts and metrics checklist
- Lesson notes
- Notebooks
- Visual guide
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 16 — Responsible AI and Bias Audit of a Chatbot

Project Question

How can we test free chatbots for bias and unfair outputs in a structured, ethical way?

What you have to do

1. Choose the target audience (CSE students, faculty or AI club members).
2. Research types of AI bias, fairness testing methods and known case studies using at least two AI research tools.
3. Collect and verify reliable sources (NIST AI RMF, EU AI Act summaries, MeitY AI guidance, papers via Elicit).
4. Create a research summary and a bias-testing checklist.
5. Design 30 paired test prompts (e.g. changing name, gender, region) and record outputs in a spreadsheet.
6. Score outputs on a simple rubric and chart the results.
7. Write a 1,200-word audit report with findings and mitigation ideas.
8. Create a 6-slide presentation and a 10-question quiz on responsible AI.
9. Automate one step (e.g. a script that tabulates prompt results from a CSV).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: no offensive prompts for shock value, report findings fairly, avoid stereotyping groups.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + data analysis + verification + report writing + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, Perplexity, NotebookLM, Gamma, GitHub; plus Google Sheets and Python.

Final submission

- Research report
- Bias-testing checklist
- Test prompt set
- Scored spreadsheet
- Audit report
- Quiz and answer key
- Presentation
- Script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 17 — Hallucination Benchmark: Testing Free Chatbots

Project Question

How reliable are free AI chatbots on technical facts, and how can users detect hallucinations?

What you have to do

1. Choose the target audience (CSE students and faculty who use AI for study).
2. Research hallucination types, causes and detection and cross-checking techniques using at least two AI research tools.
3. Collect and verify reliable sources (official docs, textbooks, papers via Consensus/Elicit).
4. Create a research summary and a fact-checking checklist for AI answers.
5. Create 40 verifiable CSE questions (10 each: programming, networks, security, theory).
6. Ask 3 chatbots, verify each answer against primary sources and record accuracy.
7. Chart accuracy by chatbot and topic, and classify error types.
8. Write a 1,000-word findings article and a 6-slide presentation.
9. Automate one step (e.g. a spreadsheet or Python script that computes accuracy scores).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: chatbots change often so date-stamp results, avoid claiming one tool is permanently best.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + verification + data analysis + writing + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, Perplexity, NotebookLM, Gamma; plus Google Sheets and Python.

Final submission

- Research report
- Fact-checking checklist for AI answers
- Question bank
- Scored results
- Charts
- Findings article
- Presentation
- Script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 18 — Prompt Engineering Workshop Kit for CSE Juniors

Project Question

How can AI help design a hands-on workshop that teaches prompt engineering to junior students?

What you have to do

1. Choose the target audience (first and second-year CSE students).
2. Research prompt patterns (role, task, format, few-shot, chain-of-thought, ReAct) using at least two AI research tools.
3. Collect and verify reliable sources (official prompt guides from OpenAI, Google and Anthropic, papers via Elicit).
4. Create a research summary and a prompt-quality checklist.
5. Create 20 before/after prompt examples across code, writing and study tasks.
6. Build a 2-hour workshop plan with exercises and answer discussion.
7. Create a visual prompt-anatomy infographic.
8. Create a 7-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a Notion template or script that stores and tags prompts).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: do not paste private data, teach verification alongside prompting, credit sources.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + learning design + writing + image generation + verification + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Gamma, Leonardo.ai, Notion, GitHub.

Final submission

- Research report
- Prompt-quality checklist
- Prompt examples
- Workshop plan
- Infographic
- Quiz and answer key
- Presentation
- Prompt template
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 19 — Technical Interview Preparation Coach

Project Question

How can AI act as a mock interviewer and feedback coach for CSE placement preparation?

What you have to do

1. Choose the target audience (final-year CSE students preparing for placements).
2. Research common technical and HR interview formats and question patterns using at least two AI research tools.
3. Collect and verify reliable sources (company career pages, official language docs, placement guides).
4. Create a research summary and an interview readiness checklist.
5. Create question banks for DSA, DBMS, OS, networks and HR rounds.
6. Run 5 mock interviews with AI and record feedback in a table.
7. Write model answers for 20 questions and verify technical correctness.
8. Create a 6-slide preparation roadmap and a 15-question quiz with answer key.
9. Automate one step (e.g. a daily interview-question reminder draft).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: never fake experience or fabricate answers for real interviews, check AI answers for correctness.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + role-play + technical explanation + verification + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, Perplexity, NotebookLM, Gamma, Make, Notion.

Final submission

- Research report
- Interview readiness checklist
- Question banks
- Mock interview log
- Model answers
- Quiz and answer key
- Roadmap deck
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 20 — Resume and LinkedIn Portfolio Kit for CSE Freshers

Project Question

How can AI help a fresher build a truthful, ATS-friendly resume and a strong online profile?

What you have to do

1. Choose the target audience (final-year CSE students and recent graduates).
2. Research ATS resume basics, recruiter expectations and profile-writing best practice using at least two AI research tools.
3. Collect and verify reliable sources (recruiter guides, LinkedIn help pages, career-services resources).
4. Create a research summary and a resume and profile quality checklist.
5. Write a resume in 3 versions (software, data, support roles) from the same real facts.
6. Create a LinkedIn headline, About section and 5-post introductory series.
7. Draft 3 cover letters and 3 networking messages.
8. Create a Notion portfolio page and a 5-slide self-introduction deck.
9. Automate one step (e.g. a workflow that tailors a resume summary to a pasted job description).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: never invent skills or experience, protect personal data, review for tone and accuracy.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + writing + editing + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Grammarly (free), Wordtune (free), Notion, Gamma, Zapier/Make.

Final submission

- Research report
- Resume and profile quality checklist
- Resume versions
- LinkedIn content
- Cover letters
- Portfolio page
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 21 — GitHub Portfolio and Open-Source Contribution Guide

Project Question

How can AI help students create a professional GitHub presence and make their first open-source contribution?

What you have to do

1. Choose the target audience (second and third-year CSE students).
2. Research README best practice, Git workflows and beginner-friendly open-source contribution steps using at least two AI research tools.
3. Collect and verify reliable sources (GitHub Docs, Open Source Guides, project CONTRIBUTING files).
4. Create a research summary and a repository quality checklist.
5. Clean up 3 repositories with AI-assisted READMEs, licences and issues.
6. Write a step-by-step first-contribution guide with screenshots.
7. Publish a GitHub Pages portfolio site.
8. Create a 6-slide talk and a 10-question quiz with answer key.
9. Automate one step (e.g. a GitHub Action that checks README sections).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: respect licences and contributor guidelines, do not spam maintainers with AI-generated pull requests.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + documentation + verification + web publishing + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, NotebookLM, Gamma, GitHub, GitHub Pages, Notion.

Final submission

- Research report
- Repository quality checklist
- Improved repos
- Contribution guide
- Portfolio site
- Quiz and answer key
- Presentation
- GitHub Action
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 22 — AI-Assisted Code Review and Documentation Toolkit

Project Question

How can AI help review code for readability, bugs and documentation while keeping humans accountable?

What you have to do

1. Choose the target audience (student project teams and open-source beginners).
2. Research code-review best practice, common bug patterns and documentation standards using at least two AI research tools.
3. Collect and verify reliable sources (Google engineering practices, PEP 8, official language docs).
4. Create a research summary and a code-review checklist.
5. Choose a small project and run AI review on 10 files, recording suggestions and accepting or rejecting each with reasons.
6. Generate docstrings, README and API docs, then verify them against the code.
7. Create a before/after code-quality table.
8. Create a 6-slide presentation and a 10-question quiz.
9. Automate one step (e.g. a script that sends a diff to a prompt template and saves the review).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: no proprietary code in prompts, human sign-off on all changes, record AI-introduced errors.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + code explanation + documentation + verification + data analysis + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub; plus VS Code and Python.

Final submission

- Research report
- Code-review checklist
- Review log
- Generated docs
- Quality table
- Quiz and answer key
- Presentation
- Script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 23 — Software Testing and Test-Case Generation Kit

Project Question

How can AI help create thorough, verified test cases and bug reports for a small application?

What you have to do

1. Choose the target audience (third-year CSE students and project teams).
2. Research test levels, equivalence partitioning, boundary-value analysis and test automation basics using at least two AI research tools.
3. Collect and verify reliable sources (ISTQB free syllabus, pytest docs, official testing guides).
4. Create a research summary and a testing checklist.
5. Pick a small app (e.g. calculator or login form) and write requirements.
6. Generate 40 test cases (functional, boundary, negative) and mark which AI missed.
7. Write automated tests with pytest and run them.
8. Create sample bug reports and a 6-slide presentation with a 10-question quiz.
9. Automate one step (e.g. a script that converts a test-case table into pytest stubs).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: verify generated tests actually fail on bugs, do not test systems you do not own.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + test design + code verification + documentation + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub; plus Python and pytest.

Final submission

- Research report
- Testing checklist
- Requirements
- Test-case suite
- Automated tests
- Bug reports
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 24 — Agile and Scrum Toolkit for Student Teams

Project Question

How can AI help a student team plan, track and review a project using Agile practices?

What you have to do

1. Choose the target audience (final-year project teams of 3 to 5 students).
2. Research Scrum roles, events, user stories and backlog management using at least two AI research tools.
3. Collect and verify reliable sources (Scrum Guide, Atlassian agile guides, Agile Manifesto).
4. Create a research summary and a sprint-planning and retrospective checklist.
5. Write user stories with acceptance criteria for a sample project.
6. Build a 3-sprint plan on a Notion board.
7. Generate retrospective questions and a sample sprint review report.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a workflow that turns meeting notes into action items).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: AI estimates are suggestions only, do not share confidential team data.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + planning + writing + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, NotebookLM, Notion, Gamma, Zapier/Make, GitHub Projects.

Final submission

- Research report
- Sprint-planning and retrospective checklist
- User stories
- Sprint board
- Retrospective report
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 25 — Beginner Web Development Starter with Accessibility

Project Question

How can AI help beginners build a small accessible website and learn HTML, CSS and JavaScript?

What you have to do

1. Choose the target audience (first and second-year CSE students).
2. Research HTML, CSS, JavaScript basics and WCAG accessibility principles using at least two AI research tools.
3. Collect and verify reliable sources (MDN Web Docs, W3C WAI and WCAG quick reference).
4. Create a research summary and an accessibility and web basics checklist.
5. Build a 3-page portfolio site with AI help and publish on GitHub Pages.
6. Write a 900-word guide on semantic HTML and responsive layout.
7. Run a free accessibility check (WAVE or Lighthouse) and record fixes.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a script that checks pages for missing alt text).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: verify generated code, credit templates, never copy sites you do not have rights to.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + code generation + documentation + verification + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, Gamma, Leonardo.ai, GitHub Pages; plus VS Code, Lighthouse and WAVE.

Final submission

- Research report
- Accessibility and web basics checklist
- Website
- Guide
- Accessibility fix log
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 26 — Web Accessibility Audit of a College or NGO Website

Project Question

How can AI help find and explain accessibility barriers on a real public website responsibly?

What you have to do

1. Choose the target audience (a college, NGO or local-business website (public pages only)).
2. Research WCAG 2.2 principles and common barriers for screen-reader, colour-blind and keyboard users using at least two AI research tools.
3. Collect and verify reliable sources (W3C WAI, GIGW guidelines, Lighthouse and WAVE docs).
4. Create a research summary and a WCAG audit checklist.
5. Audit 5 pages with free tools and manual keyboard checks.
6. Use AI to explain each issue and suggest fixes, then verify with the standard.
7. Write a 1,000-word audit report with priorities.
8. Create a 6-slide presentation for the site owner and a 10-question quiz.
9. Automate one step (e.g. a spreadsheet or script that sorts issues by severity).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: audit only public pages, no intrusive scanning, share findings constructively with the owner.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + verification + data analysis + report writing + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Gamma; plus Lighthouse, WAVE, axe DevTools (free) and Google Sheets.

Final submission

- Research report
- WCAG audit checklist
- Audit sheet
- Issue explanations
- Audit report
- Quiz and answer key
- Presentation
- Script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 27 — UI/UX Research and Prototype for a Campus App

Project Question

How can AI support user research and rapid prototyping for a campus-services app concept?

What you have to do

1. Choose the target audience (students, staff and visitors of a campus).
2. Research UX research methods, usability heuristics and persona design using at least two AI research tools.
3. Collect and verify reliable sources (Nielsen Norman Group free articles, Material Design and W3C guidance).
4. Create a research summary and a usability heuristics checklist.
5. Write a survey and interview guide, run it with 5 volunteers and summarise (anonymised).
6. Create 3 personas and a user-journey map.
7. Generate wireframes or a clickable HTML prototype and test it with users.
8. Create a 7-slide design-case deck and a 10-question quiz.
9. Automate one step (e.g. a workflow that clusters survey answers into themes).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: get consent for research, anonymise data, label AI-generated personas as fictional.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + data analysis + image generation + prototyping + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, NotebookLM, Leonardo.ai, Gamma, Notion, Make, GitHub Pages; plus Google Forms.

Final submission

- Research report
- Usability heuristics checklist
- Survey and summary
- Personas and journey
- Prototype
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 28 — Mobile App Idea Validation and Market Analysis

Project Question

How can AI help validate a mobile app idea with real market and competitor research?

What you have to do

1. Choose the target audience (student entrepreneurs and app-development teams).
2. Research market size signals, competitor features and user pain points using at least two AI research tools.
3. Collect and verify reliable sources (Play Store and App Store public listings, government statistics, published reports, Consensus and Elicit).
4. Create a research summary and an idea-validation checklist.
5. Compare 5 competitor apps in a feature table.
6. Write a 1,500-word market analysis with cited sources.
7. Create a lean canvas and 3 user stories.
8. Create a 7-slide deck and a 10-question quiz on validation concepts.
9. Automate one step (e.g. a workflow that collects competitor review themes into a sheet).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: cite all statistics, avoid scraping against terms of service, flag AI-invented numbers.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + data analysis + writing + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, Consensus, Elicit, NotebookLM, Gamma, Notion, Make.

Final submission

- Research report
- Idea-validation checklist
- Competitor table
- Market analysis
- Lean canvas
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 29 — Startup Pitch Deck and Market Analysis for a CSE Idea

Project Question

How can AI help turn a software idea into a credible pitch with verified market research?

What you have to do

1. Choose the target audience (student startup teams and innovation-cell members).
2. Research problem validation, target customers, business models and competitors using at least two AI research tools.
3. Collect and verify reliable sources (government and industry reports, public company filings, peer-reviewed papers).
4. Create a research summary and a pitch-readiness checklist.
5. Write the problem statement, solution and business model in plain language.
6. Write an 8-page market analysis with verified sources.
7. Create a 5-slide pitch deck in Gamma and a 2-minute explainer video or avatar presentation.
8. Prepare a Q&A sheet of 15 investor questions with answers.
9. Automate one step (e.g. a workflow that creates a weekly investor-update draft).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: never fabricate traction or financials, cite market data, mark all projections as assumptions.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + verification + presentation + video/audio + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, Consensus, NotebookLM, Gamma, HeyGen (free), ElevenLabs (free), Notion.

Final submission

- Research report
- Pitch-readiness checklist
- Problem and solution
- Market analysis
- Pitch deck
- Explainer video
- Q&A sheet
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 30 — Cloud Computing Basics and Free-Tier Cost Awareness

Project Question

How can AI help beginners understand cloud services and avoid surprise bills on free tiers?

What you have to do

1. Choose the target audience (third-year CSE students starting cloud labs).
2. Research IaaS, PaaS, SaaS, regions, free-tier limits and billing alerts using at least two AI research tools.
3. Collect and verify reliable sources (official AWS, Azure and Google Cloud documentation, NIST cloud definition).
4. Create a research summary and a free-tier safety checklist.
5. Write a 1,000-word beginner explainer with analogies.
6. Create a service-comparison table for compute, storage and database across providers.
7. Deploy one tiny app on a free tier or simulated environment and document steps.
8. Create a 6-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a reminder workflow for budget and free-tier expiry checks).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: set billing alerts, never share keys, verify pricing from official pages only.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + technical explanation + verification + documentation + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Gamma, Zapier/Make, GitHub; plus a cloud provider free tier or local emulator.

Final submission

- Research report
- Free-tier safety checklist
- Explainer
- Comparison table
- Deployment notes
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 31 — DevOps and Git Workflow Primer with CI Basics

Project Question

How can AI help explain Git branching, pull requests and simple CI pipelines to student teams?

What you have to do

1. Choose the target audience (second and third-year CSE project teams).
2. Research Git workflows, branching strategies and continuous integration using at least two AI research tools.
3. Collect and verify reliable sources (Pro Git book, GitHub Docs, official CI guides).
4. Create a research summary and a Git and CI checklist.
5. Write a Git command guide with a team workflow diagram.
6. Create a sample repository with branch protection rules and a pull-request template.
7. Add a GitHub Actions workflow that runs tests and linting.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. the CI workflow itself or a commit-message check).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: never commit secrets, verify AI-generated YAML, credit templates.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + technical explanation + code verification + documentation + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub and GitHub Actions.

Final submission

- Research report
- Git and CI checklist
- Git guide
- Sample repo
- CI workflow
- Quiz and answer key
- Presentation
- Demo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 32 — Smart Campus IoT Concept and Simulation

Project Question

How can AI help design and simulate a smart-campus IoT idea (energy, attendance or air quality) without needing hardware?

What you have to do

1. Choose the target audience (college administration and student innovation teams).
2. Research IoT architecture, sensors, protocols (MQTT) and IoT security risks using at least two AI research tools.
3. Collect and verify reliable sources (OWASP IoT project, NIST IoT guidance, official MQTT docs).
4. Create a research summary and an IoT design and security checklist.
5. Write a problem statement and architecture for one use case.
6. Simulate sensor data in Python and chart the results.
7. Create a system diagram and a security-risk table.
8. Create a 6-slide pitch and a 10-question quiz.
9. Automate one step (e.g. a script that turns simulated readings into a daily summary).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: simulated data only, discuss privacy and surveillance risks, no live-device attacks.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + data simulation + diagram design + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, Perplexity, NotebookLM, Gamma, Leonardo.ai, GitHub; plus Python and Wokwi simulator.

Final submission

- Research report
- IoT design and security checklist
- Architecture
- Simulation code
- Diagram and risk table
- Quiz and answer key
- Presentation
- Script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 33 — Blockchain and Crypto Scam Awareness Guide

Project Question

How can AI help explain blockchain basics and teach people to recognise crypto and investment scams?

What you have to do

1. Choose the target audience (college students and young investors).
2. Research blockchain fundamentals and common crypto/investment scam patterns using at least two AI research tools.
3. Collect and verify reliable sources (RBI, SEBI, CERT-In, FTC and CISA public advisories).
4. Create a research summary and a scam red-flag checklist.
5. Write a 1,000-word explainer on blockchain and wallets in plain language.
6. Write 6 fictional scam messages labelled "training samples" (no real tokens or platforms).
7. Create 5 social-media posts and an infographic of red flags.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. weekly scam-alert tip drafts from the checklist).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: no financial advice, no promotion of any coin, no real brands or wallet addresses.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + image generation + verification + quiz design + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Leonardo.ai, Gamma, Zapier/Make.

Final submission

- Research report
- Scam red-flag checklist
- Explainer
- Scam samples
- Posts and infographic
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 34 — Cryptography Explained for Beginners

Project Question

How can AI help students understand encryption, hashing and digital signatures through hands-on examples?

What you have to do

1. Choose the target audience (second and third-year CSE students).
2. Research symmetric vs asymmetric encryption, hashing, TLS and digital signatures using at least two AI research tools.
3. Collect and verify reliable sources (NIST publications, official OpenSSL and Python cryptography docs).
4. Create a research summary and a crypto concepts checklist.
5. Explain 6 concepts with analogies (e.g. lock and key, sealed envelope).
6. Use CyberChef to demonstrate encoding, hashing and simple ciphers on sample text.
7. Write Python examples for hashing and AES using a trusted library (never custom crypto).
8. Create a 6-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a script that compares hash outputs for sample inputs).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: use only dummy data, never design your own algorithm for real use, verify AI crypto advice with NIST.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + technical explanation + code verification + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub; plus CyberChef and Python.

Final submission

- Research report
- Crypto concepts checklist
- Concept notes
- CyberChef demos
- Python examples
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 35 — Literature Review: AI in Cybersecurity

Project Question

What does current research say about the benefits and risks of using AI for cybersecurity?

What you have to do

1. Choose the target audience (CSE final-year students or faculty).
2. Research AI for threat detection, phishing detection, and adversarial risks using at least two AI research tools.
3. Collect and verify reliable sources (peer-reviewed papers via Consensus and Elicit, arXiv, IEEE and ACM open access, NIST).
4. Create a research summary and a source-quality and bias checklist.
5. Collect 15 to 20 papers and record them in a literature matrix.
6. Write an 8–10 page literature review with themes and research gaps.
7. Create 3 charts or diagrams summarising trends.
8. Create a 7-slide summary deck and a 10-question quiz.
9. Automate one step (e.g. a workflow that turns a paper abstract into a matrix row draft).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: open-access sources only, no fabricated citations, proper referencing, disclose AI use.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + verification + data visualisation + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Perplexity, Consensus, Elicit, NotebookLM, ChatGPT, Gemini, Claude, Gamma, Notion, Obsidian.

Final submission

- Research report
- Source-quality and bias checklist
- Literature matrix
- Literature review
- Charts
- Quiz and answer key
- Presentation
- Reference list
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 36 — Green Computing and Energy-Efficient Software Awareness

Project Question

How can AI help teach students to write and run software in a more energy-efficient way?

What you have to do

1. Choose the target audience (CSE students and campus IT staff).
2. Research energy use of data centres, efficient code and sustainable practices using at least two AI research tools.
3. Collect and verify reliable sources (IEA, Green Software Foundation, peer-reviewed papers, government reports).
4. Create a research summary and a green-software checklist.
5. Write a 1,000-word article on carbon impact of computing with verified statistics.
6. Compare 2 code approaches for runtime and estimated energy.
7. Create an infographic of 10 green computing habits.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a script that logs runtime for code variants).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: cite statistics, avoid greenwashing claims, state measurement limits.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + code analysis + verification + image generation + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, Consensus, NotebookLM, Leonardo.ai, Gamma; plus Python.

Final submission

- Research report
- Green-software checklist
- Article
- Code comparison
- Infographic
- Quiz and answer key
- Presentation
- Script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 37 — Campus Helpdesk AI Agent (No-Code)

Project Question

How can a no-code AI workflow answer common campus questions and draft replies for a human to approve?

What you have to do

1. Choose the target audience (college office, library or IT helpdesk staff).
2. Research common helpdesk questions and no-code automation patterns using at least two AI research tools.
3. Collect and verify reliable sources (college public notices, Zapier/Make docs, Notion docs).
4. Create a research summary and a helpdesk FAQ and escalation checklist.
5. Build an FAQ knowledge base in Notion with at least 30 fictional or public questions.
6. Build a workflow: summarise incoming message, save to Notion, create a reply draft.
7. Write a user guide and a fallback/escalation policy.
8. Create a 6-slide demo deck and a 10-question quiz.
9. Automate one step (the Zapier/Make workflow itself, with a human-approval step).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: no private student data, humans approve replies, state limits of AI answers.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + knowledge base design + automation + verification + documentation + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Notion, Zapier (free), Make (free), NotebookLM, Gamma.

Final submission

- Research report
- Helpdesk FAQ and escalation checklist
- FAQ base
- Workflow
- User guide
- Quiz and answer key
- Presentation
- Demo video
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 38 — Tech News and Research Digest Automation

Project Question

How can AI and no-code tools create a trustworthy weekly digest of technology news and research?

What you have to do

1. Choose the target audience (CSE students, clubs and faculty).
2. Research reliable tech news and paper sources and summarisation quality using at least two AI research tools.
3. Collect and verify reliable sources (official blogs, arXiv, journal sites, press releases).
4. Create a research summary and a digest source-quality checklist.
5. Choose 10 trusted sources and record why each is trusted.
6. Create a weekly digest template with summary, source link and why it matters.
7. Produce 4 weekly editions and test summaries against originals.
8. Create a 5-slide deck and a short audio version using ElevenLabs.
9. Automate one step (the weekly digest workflow, from collecting links to a draft email).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: summarise rather than copy, always link to the original, label AI-written summaries.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + summarisation + verification + audio + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Perplexity, NotebookLM, ChatGPT, Gemini, Claude, ElevenLabs (free), Zapier/Make, Notion, Gamma.

Final submission

- Research report
- Digest source-quality checklist
- Source list
- Digest template
- 4 editions
- Audio digest
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 39 — Final-Year Project Planner and Report Assistant

Project Question

How can AI help a student team choose a project idea, plan it, and write clear technical documentation?

What you have to do

1. Choose the target audience (final-year CSE project teams).
2. Research project-selection criteria, feasibility analysis, report structure and plagiarism-safe writing using at least two AI research tools.
3. Collect and verify reliable sources (university project guidelines, IEEE templates, public project repositories).
4. Create a research summary and a project readiness and report-quality checklist.
5. Generate and shortlist 10 project ideas with a feasibility scorecard.
6. Create a project plan with milestones and risk register.
7. Draft report chapters (abstract, literature survey, design, testing) from your own notes and edit them.
8. Create a 7-slide review deck and a 10-question viva practice quiz.
9. Automate one step (e.g. a workflow that turns weekly progress notes into a status report).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: do not submit AI-written text as your own work, disclose AI help, check plagiarism and citations.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + planning + writing + verification + data analysis + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, Consensus, NotebookLM, Notion, Obsidian, Gamma, GitHub.

Final submission

- Research report
- Project readiness and report-quality checklist
- Idea scorecard
- Project plan
- Report draft
- Viva quiz
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 40 — AI Study Companion for Engineering Exams

Project Question

How can AI help a student build a personalised, accurate study system for a difficult engineering subject?

What you have to do

1. Choose the target audience (second or third-year CSE students preparing for semester exams).
2. Research effective study techniques (active recall, spaced repetition) and the chosen subject's syllabus using at least two AI research tools.
3. Collect and verify reliable sources (university syllabus, NPTEL/MIT OpenCourseWare notes, standard textbooks, learning-science papers via Consensus and Elicit).
4. Create a research summary and an exam-readiness checklist.
5. Convert one subject's syllabus into a 4-week revision plan with daily tasks.
6. Create concise notes and 40 flashcards for 5 units, then verify each against the textbook.
7. Write 20 practice questions with model answers and mark every AI error you find.
8. Create a 6-slide study-strategy deck and a 15-question quiz with answer key.
9. Automate one step (e.g. a daily revision reminder drafted from the plan).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: do not use AI to cheat in exams, verify all technical answers, protect copyrighted textbook content.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + summarisation + verification + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, NotebookLM, Perplexity, Gamma, Notion, Make.

Final submission

- Research report
- Exam-readiness checklist
- Revision plan
- Notes and flashcards
- Practice questions
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 41 — Smart Notes and Study Guide Builder with NotebookLM

Project Question

How well can NotebookLM turn lecture material into study guides, summaries and audio overviews?

What you have to do

1. Choose the target audience (college students with heavy reading loads).
2. Research note-taking methods and how source-grounded AI tools limit hallucination using at least two AI research tools.
3. Collect and verify reliable sources (NotebookLM help pages, learning-science papers, your own lecture PDFs).
4. Create a research summary and a note-quality and accuracy checklist.
5. Upload 5 lecture documents and generate a study guide, FAQ and timeline.
6. Generate an Audio Overview and check it line-by-line against the sources.
7. Compare NotebookLM with ChatGPT and Gemini on the same questions.
8. Create a 6-slide results deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a workflow that saves new notes into Notion with a summary).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: use only material you have rights to share, check audio overviews for errors, disclose AI use.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + summarisation + prompting + verification + audio + data analysis + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: NotebookLM, ChatGPT, Gemini, Claude, Gamma, Notion, Zapier/Make.

Final submission

- Research report
- Note-quality and accuracy checklist
- Study guide
- Audio overview check
- Tool comparison
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 42 — AI Coding Tutor for C Programming

Project Question

How can AI explain C programs, find bugs and guide learners without simply giving the answer?

What you have to do

1. Choose the target audience (first-year CSE students).
2. Research common C mistakes (pointers, memory, loops) and good tutoring prompts using at least two AI research tools.
3. Collect and verify reliable sources (C standard references, GNU C documentation, university C course notes).
4. Create a research summary and a C debugging checklist.
5. Write 15 buggy C programs with the expected bug type for each.
6. Ask AI to explain each bug using hints first, then full fixes, and rate the quality.
7. Write a 1,000-word guide to the 10 most common C beginner errors.
8. Create a 6-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a script that compiles code and stores error messages for AI explanation).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: run all code in a safe local environment, verify fixes, do not hand in AI-written solutions as your own.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + code explanation + debugging + verification + writing + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub; plus GCC and VS Code.

Final submission

- Research report
- C debugging checklist
- Buggy program set
- Tutor rating log
- Mistakes guide
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 43 — Java Object-Oriented Concepts Learning Pack

Project Question

How can AI help teach OOP ideas (classes, inheritance, polymorphism) with clear examples and exercises?

What you have to do

1. Choose the target audience (second-year CSE students).
2. Research OOP principles and common misconceptions in Java using at least two AI research tools.
3. Collect and verify reliable sources (Oracle Java Tutorials, official Java docs, university notes).
4. Create a research summary and an OOP concepts checklist.
5. Explain 8 OOP concepts with real-life analogies.
6. Write and test 10 Java examples with UML class diagrams.
7. Create 20 exercises with solutions and verify by running them.
8. Create a 7-slide deck and a 15-question quiz with answer key.
9. Automate one step (e.g. a script that generates exercise variations from a template).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: verify all code, credit analogies from sources, do not submit unexplained AI code.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + technical explanation + code verification + diagram design + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub; plus JDK and VS Code.

Final submission

- Research report
- OOP concepts checklist
- Concept notes
- Tested code
- UML diagrams
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 44 — Competitive Programming Starter Guide

Project Question

How can AI help beginners learn problem-solving patterns without becoming dependent on it?

What you have to do

1. Choose the target audience (second and third-year CSE students).
2. Research common problem patterns (two pointers, sliding window, DP) and practice strategy using at least two AI research tools.
3. Collect and verify reliable sources (Codeforces/LeetCode public editorials, CP-Algorithms, official language docs).
4. Create a research summary and a problem-solving approach checklist.
5. Select 15 problems across 5 patterns and write your own solution approach first.
6. Compare your approach with AI hints and note where AI was wrong or misleading.
7. Write a 1,000-word pattern guide with complexity analysis.
8. Create a 6-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a tracker that logs problem, pattern, time taken and result).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: follow contest rules, never use AI during live contests, verify complexity claims.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + problem solving + code verification + data analysis + writing + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, Notion, GitHub; plus Python/C++ and Google Sheets.

Final submission

- Research report
- Problem-solving approach checklist
- Problem log
- Approach comparison
- Pattern guide
- Quiz and answer key
- Presentation
- Tracker
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 45 — Exploratory Data Analysis on an Open Dataset

Project Question

How can AI assist the full data-analysis workflow while the analyst checks every number?

What you have to do

1. Choose the target audience (CSE or data-science students).
2. Research EDA steps, data cleaning and common statistical pitfalls using at least two AI research tools.
3. Collect and verify reliable sources (pandas and matplotlib docs, data.gov.in or Kaggle dataset documentation).
4. Create a research summary and an EDA and data-quality checklist.
5. Pick an open dataset and document its source, licence and limits.
6. Clean and analyse it in Python with AI help, verifying each result by hand or a second method.
7. Create 5 charts and a 1,000-word insights report.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a reusable notebook that generates a data-quality summary).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: open data only, no personal data, state limitations and avoid causal claims without evidence.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + data analysis + code verification + visualisation + writing + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub; plus Python, pandas and Google Colab.

Final submission

- Research report
- EDA and data-quality checklist
- Dataset note
- Analysis notebook
- Charts
- Insights report
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 46 — Data Storytelling with Open Government Data

Project Question

How can AI help turn public data into a clear, honest story that ordinary readers understand?

What you have to do

1. Choose the target audience (general public and local-community readers).
2. Research data visualisation principles and misleading-chart pitfalls using at least two AI research tools.
3. Collect and verify reliable sources (data.gov.in, census and ministry open data, Tufte-style visual guidance).
4. Create a research summary and a chart honesty checklist.
5. Choose one public issue (e.g. transport, education) and collect 2–3 verified datasets.
6. Create 4 charts and annotate them with plain-language takeaways.
7. Write a 900-word data story and a 5-post social series.
8. Create a 6-slide deck and a 10-question quiz on reading charts.
9. Automate one step (e.g. a script that refreshes charts when the CSV is updated).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: cite every dataset, avoid cherry-picking, state data limits clearly.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + data analysis + visualisation + writing + presentation + verification + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Leonardo.ai, Gamma; plus Google Sheets and Python.

Final submission

- Research report
- Chart honesty checklist
- Dataset list
- Charts
- Data story
- Social posts
- Presentation
- Script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 47 — SQL vs NoSQL Decision Guide

Project Question

How can AI help students compare relational and NoSQL databases and choose one for a real use case?

What you have to do

1. Choose the target audience (third-year CSE students).
2. Research relational, document, key-value and graph databases, CAP theorem and trade-offs using at least two AI research tools.
3. Collect and verify reliable sources (official PostgreSQL, MongoDB and Redis documentation).
4. Create a research summary and a database selection checklist.
5. Describe 4 use cases (e-commerce, chat, analytics, social graph).
6. Build tiny prototypes in SQLite and a document-style store for one use case.
7. Write a 1,000-word decision guide with a comparison table.
8. Create a 6-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a script that loads sample data into both databases).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: verify all benchmark claims, fictional data only, avoid vendor bias.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + technical explanation + code verification + writing + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, Perplexity, NotebookLM, Gamma, GitHub; plus SQLite and Python.

Final submission

- Research report
- Database selection checklist
- Use-case set
- Prototypes
- Decision guide
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 48 — REST API Design and Documentation Kit

Project Question

How can AI help design, test and document a clean REST API for a small application?

What you have to do

1. Choose the target audience (third-year CSE students).
2. Research REST principles, HTTP status codes, versioning and OpenAPI documentation using at least two AI research tools.
3. Collect and verify reliable sources (MDN HTTP docs, OpenAPI specification, official framework docs).
4. Create a research summary and an API design checklist.
5. Design a small API (e.g. library or to-do) with endpoint tables.
6. Implement it in Flask or FastAPI and test with Postman or curl.
7. Generate OpenAPI documentation and verify it matches the code.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a script that checks endpoints against the documentation).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: no real user data, verify AI-generated code, never expose keys.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + code generation + verification + documentation + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub; plus Python, FastAPI and Postman.

Final submission

- Research report
- API design checklist
- Endpoint design
- Working API
- OpenAPI docs
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 49 — No-Code FAQ Chatbot for College Admissions

Project Question

How can a no-code AI assistant answer admission questions accurately and hand off hard ones to a human?

What you have to do

1. Choose the target audience (prospective students and the admissions office).
2. Research common admission questions and chatbot design principles using at least two AI research tools.
3. Collect and verify reliable sources (public college brochure and notices, official admission guidelines).
4. Create a research summary and a chatbot quality and escalation checklist.
5. Build a 40-question verified FAQ from public documents.
6. Create the chatbot flow (Gemini or Claude project, or Notion plus Make) with fallback replies.
7. Test with 30 user questions and record accuracy and failures.
8. Create a 6-slide demo deck and a 10-question quiz.
9. Automate one step (the automation that logs unanswered questions into Notion).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: use only public information, never promise admission outcomes, show a human-contact fallback.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + knowledge base design + automation + verification + data analysis + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, NotebookLM, Notion, Zapier/Make, Gamma.

Final submission

- Research report
- Chatbot quality and escalation checklist
- FAQ base
- Chatbot flow
- Test results
- Quiz and answer key
- Presentation
- Demo video
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 50 — Sentiment Analysis on Product Reviews (NLP Basics)

Project Question

How can students build and honestly evaluate a simple sentiment classifier using open data?

What you have to do

1. Choose the target audience (third-year CSE students new to NLP).
2. Research text preprocessing, sentiment classification and evaluation metrics using at least two AI research tools.
3. Collect and verify reliable sources (scikit-learn docs, Hugging Face docs, open review datasets).
4. Create a research summary and an NLP project checklist.
5. Select an open review dataset and document its licence.
6. Train a baseline model and compare it with a pre-trained Hugging Face model.
7. Analyse errors (sarcasm, negation) and chart metrics.
8. Create a 7-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a script that exports predictions and errors into a table).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: open data only, report bias and limits, do not claim production accuracy.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + code verification + data analysis + visualisation + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, Hugging Face, NotebookLM, Gamma, GitHub; plus Python and Google Colab.

Final submission

- Research report
- NLP project checklist
- Dataset note
- Models
- Error analysis
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 51 — Image Classification Mini-Project with Hugging Face

Project Question

How can a beginner build a small image classifier and understand where it fails?

What you have to do

1. Choose the target audience (third-year CSE students).
2. Research image classification basics, transfer learning and dataset bias using at least two AI research tools.
3. Collect and verify reliable sources (Hugging Face docs, TensorFlow/PyTorch tutorials, open image datasets).
4. Create a research summary and a vision-project checklist.
5. Choose an open image dataset (e.g. flowers or digits) and document it.
6. Fine-tune or test a pre-trained model in Colab.
7. Test on 20 of your own safe images and record failures.
8. Create a 7-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a script that creates a confusion matrix report).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: no face recognition on real people, open data only, report limits honestly.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + code verification + data analysis + visualisation + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, Hugging Face, NotebookLM, Gamma, GitHub; plus Python and Google Colab.

Final submission

- Research report
- Vision-project checklist
- Dataset note
- Notebook
- Failure analysis
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 52 — Explainable AI for Beginners

Project Question

How can we explain why a simple ML model makes its predictions in terms a non-expert can follow?

What you have to do

1. Choose the target audience (CSE students and non-technical stakeholders).
2. Research explainability methods (feature importance, SHAP, LIME) and their limits using at least two AI research tools.
3. Collect and verify reliable sources (scikit-learn docs, SHAP and LIME documentation, papers via Elicit).
4. Create a research summary and an explainability communication checklist.
5. Train a simple model on an open dataset.
6. Generate feature-importance and explanation plots.
7. Write a 1,000-word plain-language explainer, checked by one non-technical reader.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a notebook that outputs a one-page explanation report).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: explanations are approximations, avoid overclaiming fairness, open data only.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + code verification + visualisation + writing + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, Elicit, NotebookLM, Gamma, GitHub; plus Python and Google Colab.

Final submission

- Research report
- Explainability communication checklist
- Model notebook
- Explanation plots
- Explainer article
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 53 — AI Regulation Comparison: India, EU and US

Project Question

How do AI rules and guidance differ across India, the EU and the US, and what do they mean for students and developers?

What you have to do

1. Choose the target audience (CSE students, startups and faculty).
2. Research the EU AI Act risk categories, US executive and state-level actions, and India's AI governance guidance using at least two AI research tools.
3. Collect and verify reliable sources (official EU, US government and MeitY documents, NIST AI RMF, law-firm summaries for cross-checking).
4. Create a research summary and a regulation-summary accuracy checklist.
5. Build a comparison table across risk approach, enforcement and penalties.
6. Write a 1,500-word briefing citing official documents.
7. Create 3 short case studies showing how each rule applies to a project.
8. Create a 7-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. an alert workflow that tracks new AI-policy news into a sheet).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: laws change quickly, always date-stamp, AI summaries are not legal advice.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + summarisation + verification + writing + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Perplexity, Consensus, NotebookLM, ChatGPT, Gemini, Claude, Gamma, Zapier/Make, Notion.

Final submission

- Research report
- Regulation-summary accuracy checklist
- Comparison table
- Briefing
- Case studies
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 54 — AI and Jobs: Skills Gap Analysis for CSE Graduates

Project Question

Which skills do entry-level CSE job descriptions demand, and how is AI changing them?

What you have to do

1. Choose the target audience (final-year CSE students).
2. Research job-market skill trends and AI's effect on entry-level roles using at least two AI research tools.
3. Collect and verify reliable sources (public job postings, government labour statistics, industry skill reports).
4. Create a research summary and a skills-gap self-assessment checklist.
5. Collect 30 public job descriptions and extract required skills into a sheet.
6. Use AI to cluster skills, then check the clusters manually.
7. Write an 8-page skills-gap analysis with a personal upskilling plan.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a workflow that adds a new job posting's skills to the sheet).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: cite statistics, avoid scraping against terms of service, avoid fear-mongering claims.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + data analysis + verification + writing + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Perplexity, Consensus, Elicit, ChatGPT, Gemini, Claude, NotebookLM, Gamma, Notion; plus Google Sheets.

Final submission

- Research report
- Skills-gap self-assessment checklist
- Skills sheet
- Cluster analysis
- Gap report
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 55 — Copyright and Licensing Guide for AI-Generated Content and Code

Project Question

What can students safely do with AI-generated text, images and code under current copyright and licence rules?

What you have to do

1. Choose the target audience (CSE students, designers and open-source beginners).
2. Research copyright basics, open-source licences, AI training-data disputes and tool terms of use using at least two AI research tools.
3. Collect and verify reliable sources (official licence texts, Creative Commons, court and government publications, tool terms pages).
4. Create a research summary and a safe-use and attribution checklist.
5. Write a plain-language guide on copyright vs licences (MIT, GPL, CC).
6. Check the terms of 5 free AI tools for commercial use and ownership of outputs.
7. Create 6 scenario cards (e.g. selling AI images, copying code) with safe answers.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a licence-check reminder for new project dependencies).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: not legal advice, rules differ by country and change often, date-stamp every claim.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + summarisation + verification + writing + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Perplexity, Consensus, NotebookLM, ChatGPT, Gemini, Claude, Gamma, Notion, GitHub.

Final submission

- Research report
- Safe-use and attribution checklist
- Licence guide
- Terms comparison
- Scenario cards
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 56 — Open-Source LLM Exploration and Comparison

Project Question

How do open-source language models compare with free hosted chatbots for student tasks?

What you have to do

1. Choose the target audience (CSE students and AI club members).
2. Research open-source model families, model cards, licences and evaluation basics using at least two AI research tools.
3. Collect and verify reliable sources (Hugging Face model cards, official model licences, published benchmarks).
4. Create a research summary and a model-evaluation checklist.
5. Select 3 open models and record size, licence and intended use.
6. Run 20 identical tasks (summary, code, reasoning) on each model or hosted demo.
7. Score outputs with a rubric and chart the results.
8. Create a 7-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a notebook that sends a prompt set to each model and saves outputs).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: respect licences, report limits, date-stamp results.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + data analysis + verification + visualisation + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Hugging Face, ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub; plus Google Colab.

Final submission

- Research report
- Model-evaluation checklist
- Model table
- Task results
- Charts
- Quiz and answer key
- Presentation
- Notebook
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 57 — Multimodal AI Tools Comparison for Students

Project Question

How well do free multimodal AI tools handle images, documents and audio for study tasks?

What you have to do

1. Choose the target audience (college students).
2. Research multimodal capabilities and common failure modes using at least two AI research tools.
3. Collect and verify reliable sources (official product pages and documentation, independent evaluations).
4. Create a research summary and a multimodal output checklist.
5. Design 15 tasks (read a diagram, summarise a PDF, describe an image, transcribe audio).
6. Run them across 3 free tools and record accuracy.
7. Write a 1,000-word comparison with examples of failures.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a results sheet that auto-calculates scores).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: no private images or documents, date-stamp results, avoid overclaiming.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + verification + data analysis + writing + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Copilot, NotebookLM, Gamma; plus Google Sheets.

Final submission

- Research report
- Multimodal output checklist
- Task set
- Score sheet
- Comparison article
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 58 — Multi-Agent Research Assistant Demo

Project Question

How can several AI agents with different roles work together on a research task, and where do they go wrong?

What you have to do

1. Choose the target audience (CSE students and AI club members).
2. Research agent design, tool use and multi-agent patterns (CrewAI / AutoGen style) using at least two AI research tools.
3. Collect and verify reliable sources (official CrewAI and AutoGen documentation, agent-design guides).
4. Create a research summary and an agent-reliability checklist.
5. Design 3 agents (researcher, writer, reviewer) with clear roles.
6. Run a demo, or simulate the same roles manually with ChatGPT, Gemini and Claude.
7. Record failures, loops and wrong claims, and compare with a single-prompt result.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (the agent pipeline itself, kept to demo level).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: human review of all outputs, no access to private accounts, keep agents sandboxed.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + agent design + automation + verification + data analysis + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, CrewAI/AutoGen (open source), Gamma, GitHub.

Final submission

- Research report
- Agent-reliability checklist
- Agent design
- Run logs
- Failure analysis
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 59 — Personal Productivity System with Notion and Obsidian

Project Question

How can AI help build a personal knowledge and task system that a student actually keeps using?

What you have to do

1. Choose the target audience (college students and early-career developers).
2. Research personal knowledge management methods (PARA, Zettelkasten) and habit design using at least two AI research tools.
3. Collect and verify reliable sources (Notion and Obsidian official guides, productivity research).
4. Create a research summary and a weekly review checklist.
5. Build a Notion workspace for tasks, notes and goals.
6. Create an Obsidian vault with linked notes for 2 subjects.
7. Run a 2-week trial and log time saved and problems.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a weekly review summary generated from your task list).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: keep private notes out of AI prompts, report honestly if the system did not help.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + knowledge management + data analysis + writing + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Notion, Obsidian, Mem (free), ChatGPT, Gemini, Claude, Gamma, Zapier/Make.

Final submission

- Research report
- Weekly review checklist
- Notion workspace
- Obsidian vault
- Trial log
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 60 — AI-Powered Email and Calendar Routine

Project Question

How can AI cut email and scheduling time without causing mistakes or tone problems?

What you have to do

1. Choose the target audience (students, interns and working professionals).
2. Research email-triage methods and AI drafting risks using at least two AI research tools.
3. Collect and verify reliable sources (Google Workspace and Microsoft Copilot help pages, productivity guides).
4. Create a research summary and an email-quality and privacy checklist.
5. Create 10 sample inboxes of fictional messages and triage them with AI.
6. Draft replies in 3 tones and compare quality.
7. Plan a week schedule with AI and record conflicts it missed.
8. Create a 5-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a workflow that turns an email into a task and a reply draft).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: never paste real private emails, review every draft before sending, disclose AI use where required.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + writing + planning + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Copilot (free), Grammarly (free), Notion, Zapier/Make.

Final submission

- Research report
- Email-quality and privacy checklist
- Triage log
- Reply drafts
- Weekly plan
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 61 — Meeting Notes to Action Items Automation

Project Question

How reliably can AI turn raw meeting notes into accurate summaries and action items?

What you have to do

1. Choose the target audience (student project teams and clubs).
2. Research meeting-summary best practice and common AI summarisation errors using at least two AI research tools.
3. Collect and verify reliable sources (Atlassian and Google meeting guides, summarisation research).
4. Create a research summary and a meeting-notes accuracy checklist.
5. Write 5 fictional meeting transcripts.
6. Generate summaries and action items, then score accuracy against the transcripts.
7. Design a standard output template (decisions, owners, deadlines).
8. Create a 5-slide deck and a 10-question quiz with answer key.
9. Automate one step (the workflow that saves summaries and tasks into Notion).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: get consent before recording real meetings, fictional transcripts for submission.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + summarisation + verification + data analysis + automation + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, NotebookLM, Notion, Zapier/Make, Gamma.

Final submission

- Research report
- Meeting-notes accuracy checklist
- Transcripts
- Output template
- Accuracy scores
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 62 — Technical Blog Writing Series for Developers

Project Question

How can AI help a student write clear, original and accurate technical blog posts?

What you have to do

1. Choose the target audience (CSE students building a public developer profile).
2. Research technical writing best practice and developer-blog structure using at least two AI research tools.
3. Collect and verify reliable sources (Google developer documentation style guide, popular technical blogs for style analysis only).
4. Create a research summary and a technical-writing quality checklist.
5. Choose a topic series and plan 3 posts.
6. Write post 1 as a 1,000-word article with tested code.
7. Write posts 2 and 3 as shorter pieces and create 5 social posts.
8. Create a before/after comparison of AI draft versus final edit, plus a 5-slide deck.
9. Automate one step (e.g. a content calendar and draft-reminder workflow).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: original work only, test every code sample, disclose AI use, never copy other blogs.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + writing + editing + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Grammarly (free), Wordtune (free), Notion, Gamma, Make, GitHub.

Final submission

- Research report
- Technical-writing quality checklist
- Content plan
- Blog post
- Two more posts
- Social posts
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 63 — README and Documentation Kit for Student Projects

Project Question

How can AI help students document projects so others can understand, install and use them?

What you have to do

1. Choose the target audience (student developers and project teams).
2. Research good documentation structure and README templates using at least two AI research tools.
3. Collect and verify reliable sources (GitHub Docs, Make a README guide, Write the Docs).
4. Create a research summary and a documentation completeness checklist.
5. Choose 3 of your projects and write READMEs with AI help.
6. Add install guides, usage examples and contribution notes, then test them on a fresh machine or classmate.
7. Write a 900-word guide on documentation mistakes.
8. Create a 5-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a GitHub Action that checks README headings).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: only describe what the code actually does, check every command, credit templates.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + documentation + verification + writing + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, NotebookLM, Gamma, Notion, GitHub.

Final submission

- Research report
- Documentation completeness checklist
- Three READMEs
- Install test notes
- Guide
- Quiz and answer key
- Presentation
- GitHub Action
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 64 — Research Paper Reading and Summarisation Assistant

Project Question

How can AI help students read research papers faster while still catching errors and missing detail?

What you have to do

1. Choose the target audience (final-year students and early researchers).
2. Research how to read a paper and common AI summary errors using at least two AI research tools.
3. Collect and verify reliable sources (open-access papers from arXiv, PubMed Central and journals, NotebookLM and Elicit docs).
4. Create a research summary and a paper-summary verification checklist.
5. Choose 6 open-access papers on one topic.
6. Summarise each with AI, then compare the summary to the abstract, method and results.
7. Build a comparison matrix and write a 1,500-word synthesis.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a workflow that turns a paper link into a matrix row draft).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: open-access sources only, never invent citations, disclose AI use.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + summarisation + prompting + verification + writing + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Elicit, Consensus, Perplexity, NotebookLM, ChatGPT, Gemini, Claude, Gamma, Notion.

Final submission

- Research report
- Paper-summary verification checklist
- Paper list
- Summary checks
- Comparison matrix
- Synthesis
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 65 — Plagiarism-Safe Academic Writing Guide

Project Question

How can students use AI to improve writing without plagiarism or false citations?

What you have to do

1. Choose the target audience (college students writing reports and papers).
2. Research plagiarism types, citation styles and institutional AI-use policies using at least two AI research tools.
3. Collect and verify reliable sources (university integrity policies, APA/IEEE guides, publisher AI policies).
4. Create a research summary and an academic-integrity checklist.
5. Write a 1,000-word guide on safe AI use in assignments.
6. Test AI-generated citations for 20 references and record how many are fake or wrong.
7. Create 6 scenario cards (paraphrasing, paste-and-edit) with the correct action.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a citation-checking sheet with verification status).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: follow your institution's policy, never submit AI work as your own, disclose use.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + verification + writing + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Perplexity, Consensus, Elicit, ChatGPT, Gemini, Claude, Grammarly (free), Gamma, Notion.

Final submission

- Research report
- Academic-integrity checklist
- Guide
- Citation test
- Scenario cards
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 66 — Mobile App UI Design Concepts with AI Image Tools

Project Question

How can AI image tools speed up mobile app UI concept exploration, and what must still be designed by a person?

What you have to do

1. Choose the target audience (student app teams).
2. Research mobile UI principles and AI image-prompting for interface concepts using at least two AI research tools.
3. Collect and verify reliable sources (Material Design and Apple HIG public guidelines, accessibility guidance).
4. Create a research summary and a UI concept review checklist.
5. Choose an app idea and write 3 user stories.
6. Generate 15 UI concept images with Leonardo.ai or Bing Image Creator and refine prompts.
7. Pick one concept and build a simple clickable HTML mockup.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a prompt-tracking sheet with settings and results).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: no copying existing app designs, check licence and terms for generated images, label as concepts.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + image generation + design thinking + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Leonardo.ai, Bing Image Creator, Flux via Hugging Face, ChatGPT, Gemini, Claude, Gamma, GitHub Pages.

Final submission

- Research report
- UI concept review checklist
- User stories
- 15 concept images
- HTML mockup
- Quiz and answer key
- Presentation
- Prompt sheet
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 67 — Logo and Brand Identity Concepts for a Student Startup

Project Question

How can AI help create a consistent brand concept while keeping it original and legally safe?

What you have to do

1. Choose the target audience (student startups and clubs).
2. Research brand identity basics and logo-design principles using at least two AI research tools.
3. Collect and verify reliable sources (public design guides, trademark-search guidance from official registries).
4. Create a research summary and a brand consistency checklist.
5. Write a brand brief (values, audience, tone).
6. Generate 15 logo concept images and select 3 finalists with reasons.
7. Create a mini brand guide (colours, fonts, voice) and 5 sample posts.
8. Create a 6-slide brand deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a content-calendar workflow using the brand voice).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: check similarity to existing marks, disclose AI use, label as concept only.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + image generation + writing + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Leonardo.ai, Bing Image Creator, ChatGPT, Gemini, Claude, Gamma, Notion, Make.

Final submission

- Research report
- Brand consistency checklist
- Brand brief
- Logo concepts
- Brand guide
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 68 — Educational Explainer Video Series with AI Avatar

Project Question

How can AI avatar and voice tools create short, accurate explainer videos for a technical topic?

What you have to do

1. Choose the target audience (junior students and self-learners).
2. Research explainer-video structure and AI voice and avatar tool limits using at least two AI research tools.
3. Collect and verify reliable sources (HeyGen and ElevenLabs help pages, video-learning research).
4. Create a research summary and a video script accuracy checklist.
5. Choose one topic and write 3 short scripts (about 2 minutes each).
6. Produce voice or avatar versions with free tools and add slides.
7. Ask 5 viewers for feedback and record comprehension.
8. Create a 5-slide project deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a workflow that turns a script into a caption file).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: never clone anyone's voice without consent, label synthetic voices and avatars, check script accuracy.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + writing + video/audio + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, HeyGen (free), ElevenLabs (free), Gamma, Runway/Pika/Kling (free credits).

Final submission

- Research report
- Video script accuracy checklist
- Scripts
- Videos
- Feedback summary
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 69 — Audio Study Series: Podcasts from Documents

Project Question

How accurate and useful are AI-generated podcasts as a study aid?

What you have to do

1. Choose the target audience (students with long reading lists).
2. Research audio learning and AI summarisation accuracy using at least two AI research tools.
3. Collect and verify reliable sources (NotebookLM help pages, learning-research papers, your own course PDFs).
4. Create a research summary and an audio-accuracy checklist.
5. Create 4 Audio Overviews from 4 different documents.
6. Fact-check each against its source and record errors.
7. Compare audio learning with reading using a short 5-person quiz test.
8. Create a 5-slide results deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a workflow that files transcripts and error notes in Notion).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: use only documents you may share, label as AI-generated audio, check facts before relying on them.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + summarisation + audio + verification + data analysis + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: NotebookLM, ElevenLabs (free), ChatGPT, Gemini, Claude, Gamma, Notion.

Final submission

- Research report
- Audio-accuracy checklist
- Audio overviews
- Fact-check log
- Learning test
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 70 — Linux Command Line Learning Kit

Project Question

How can AI help beginners learn Linux commands safely through guided practice?

What you have to do

1. Choose the target audience (first and second-year CSE students).
2. Research core Linux commands, file permissions and safe-practice rules using at least two AI research tools.
3. Collect and verify reliable sources (GNU and Linux man pages, Ubuntu documentation).
4. Create a research summary and a command safety checklist.
5. Create a 2-week plan with daily 20-minute drills.
6. Write 30 command exercises with expected output and verify them in a VM.
7. Write a 900-word guide to dangerous commands and safe habits.
8. Create a 6-slide deck and a 15-question quiz with answer key.
9. Automate one step (e.g. a shell script that checks whether exercises were completed).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: practise only inside a VM, never run unknown commands from AI without understanding them.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + technical explanation + verification + writing + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub; plus Ubuntu in VirtualBox or WSL.

Final submission

- Research report
- Command safety checklist
- 2-week plan
- Exercise set
- Safety guide
- Quiz and answer key
- Presentation
- Shell script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 71 — Docker and Containers for Beginners

Project Question

How can AI help students understand containers and build a first reproducible Docker setup?

What you have to do

1. Choose the target audience (third-year CSE students).
2. Research containers vs virtual machines, images, Dockerfiles and Compose using at least two AI research tools.
3. Collect and verify reliable sources (official Docker documentation).
4. Create a research summary and a Docker best-practice checklist.
5. Write an explainer with a container-vs-VM diagram.
6. Containerise a small web app and write a clean Dockerfile.
7. Review AI-written Dockerfiles for security and size mistakes.
8. Create a 6-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a GitHub Action that builds the image on each push).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: never commit secrets, verify base images from official sources, run only your own code.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + technical explanation + code verification + documentation + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, GitHub; plus Docker Desktop and Python.

Final submission

- Research report
- Docker best-practice checklist
- Explainer
- Dockerised app
- Review notes
- Quiz and answer key
- Presentation
- GitHub Action
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 72 — Software Requirements Specification (SRS) Writing Kit

Project Question

How can AI help write a clear, testable SRS without inventing requirements?

What you have to do

1. Choose the target audience (third-year CSE project teams).
2. Research SRS structure (IEEE-style), functional vs non-functional requirements and traceability using at least two AI research tools.
3. Collect and verify reliable sources (IEEE 29148 summaries, university templates, public SRS examples).
4. Create a research summary and an SRS quality checklist.
5. Pick a small system (e.g. library management) and gather real requirements from a classmate or staff member.
6. Draft the SRS with AI and remove requirements nobody asked for.
7. Add use cases and a requirements-to-tests table.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a script that checks requirement IDs are unique and linked to tests).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: mark all AI-suggested requirements, get stakeholder approval, no confidential company specs.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + writing + requirements analysis + verification + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, NotebookLM, Notion, Gamma, GitHub.

Final submission

- Research report
- SRS quality checklist
- SRS document
- Use cases
- Traceability table
- Quiz and answer key
- Presentation
- Script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 73 — Software Design Patterns Visual Guide

Project Question

How can AI help explain design patterns with clear diagrams and tested code?

What you have to do

1. Choose the target audience (third-year CSE students).
2. Research creational, structural and behavioural patterns and when not to use them using at least two AI research tools.
3. Collect and verify reliable sources (Refactoring.Guru-style references, language docs, GoF summaries).
4. Create a research summary and a pattern-selection checklist.
5. Explain 8 patterns with real-life analogies.
6. Write and run code for each in Python or Java.
7. Draw UML-style diagrams for each pattern.
8. Create a 7-slide deck and a 15-question quiz with answer key.
9. Automate one step (e.g. a script that runs all pattern examples with one command).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: verify code and diagrams, do not force patterns where simple code works.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + technical explanation + code verification + diagram design + quiz design + presentation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, NotebookLM, Gamma, Leonardo.ai, GitHub; plus Python or JDK.

Final submission

- Research report
- Pattern-selection checklist
- Pattern notes
- Tested code
- Diagrams
- Quiz and answer key
- Presentation
- GitHub repo
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 74 — Defensive Guide to SQL Injection and Database Security

Project Question

How can AI help teach database security defensively without producing attack tools?

What you have to do

1. Choose the target audience (second and third-year CSE students).
2. Research SQL injection prevention, least privilege and safe query practices using at least two AI research tools.
3. Collect and verify reliable sources (OWASP, NIST, official database security docs).
4. Create a research summary and a database security checklist.
5. Build a tiny local app with a vulnerable and a fixed version (own lab only).
6. Explain each flaw and fix in plain language.
7. Write a 1,000-word defensive guide with parameterised-query examples.
8. Create a 6-slide deck and a 12-question quiz with answer key.
9. Automate one step (e.g. a script that scans your own code for string-built SQL).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: local lab only, no attacks on real systems, do not publish working attack payloads.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + code verification + writing + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, DeepSeek, Perplexity, NotebookLM, Gamma, GitHub; plus SQLite and Python.

Final submission

- Research report
- Database security checklist
- Lab app
- Flaw and fix notes
- Defensive guide
- Quiz and answer key
- Presentation
- Scan script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 75 — Safe Use of Generative AI at Work: Data Leak Prevention

Project Question

How can students and small teams use AI tools without leaking private or confidential data?

What you have to do

1. Choose the target audience (interns, startups and student teams).
2. Research data-leak risks with AI tools, tool privacy settings and acceptable-use policies using at least two AI research tools.
3. Collect and verify reliable sources (NIST AI RMF, CERT-In, tool privacy pages, company AI-policy examples).
4. Create a research summary and a do-and-don't prompt safety checklist.
5. Write a one-page acceptable-use policy for a small team.
6. Create 8 fictional scenarios (pasting client data, code, personal data) with safe alternatives.
7. Test redaction by rewriting 5 sensitive prompts into safe versions.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a reminder workflow or script that flags emails and IDs before pasting).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: fictional data only, tool privacy terms change, date-stamp claims.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + verification + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Gamma, Notion, Make; plus Python (regex checker).

Final submission

- Research report
- Do-and-don't prompt safety checklist
- Policy
- Scenario set
- Redaction tests
- Quiz and answer key
- Presentation
- Script
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 76 — Cyber Law and Digital Rights Awareness (IT Act)

Project Question

How can AI help explain key cyber laws and digital rights in plain language?

What you have to do

1. Choose the target audience (college students and the general public).
2. Research India's IT Act basics, common cyber offences and reporting routes using at least two AI research tools.
3. Collect and verify reliable sources (official IT Act text, MeitY, cybercrime.gov.in and the National Cyber Crime Portal).
4. Create a research summary and a know-your-rights and reporting checklist.
5. Write a plain-language 1,000-word guide.
6. Create 6 case scenarios (fraud, harassment, data theft) and the correct reporting steps.
7. Create an infographic: "What to do if I am a victim".
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. weekly awareness posts drafted from the guide).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: not legal advice, check section numbers against official text, laws may be amended.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + verification + image generation + quiz design + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: Perplexity, NotebookLM, ChatGPT, Gemini, Claude, Leonardo.ai, Gamma, Make.

Final submission

- Research report
- Know-your-rights and reporting checklist
- Guide
- Case scenarios
- Infographic
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 77 — Digital Literacy Workshop for Rural Students

Project Question

How can AI help prepare a simple, low-bandwidth digital literacy workshop for school students?

What you have to do

1. Choose the target audience (school students in rural or low-resource areas).
2. Research digital literacy skills, safe internet use and low-bandwidth teaching methods using at least two AI research tools.
3. Collect and verify reliable sources (NCERT, Digital India resources, UNESCO digital literacy frameworks).
4. Create a research summary and a workshop readiness checklist.
5. Plan a 2-hour workshop with activities that need little or no internet.
6. Create printable worksheets and a regional-language handout.
7. Create a poster and a short audio guide.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a workflow that turns feedback forms into an improvement summary).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: get school permission, protect children's privacy, check translations with a native speaker.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + translation + image generation + audio + quiz design + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, NotebookLM, Leonardo.ai, ElevenLabs (free), Gamma, Make.

Final submission

- Research report
- Workshop readiness checklist
- Workshop plan
- Worksheets and handout
- Poster and audio
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 78 — Assistive Technology and Accessibility Guide Using AI

Project Question

How can AI-powered captions, screen readers and text tools improve access for learners with disabilities?

What you have to do

1. Choose the target audience (students with disabilities, faculty and campus IT teams).
2. Research assistive technologies (screen readers, captions, text-to-speech) and WCAG basics using at least two AI research tools.
3. Collect and verify reliable sources (W3C WAI, accessibility guidance, official tool documentation).
4. Create a research summary and an inclusive-content checklist.
5. Test 4 free tools (e.g. live captions, text-to-speech, image description) on sample content.
6. Record accuracy and failures, especially for technical terms.
7. Write a 1,000-word practical guide for teachers and students.
8. Create a 6-slide deck and a 10-question quiz with answer key.
9. Automate one step (e.g. a workflow that generates alt-text drafts for images).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: consult users with lived experience, never claim AI replaces human accessibility review.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + verification + writing + audio + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, NotebookLM, ElevenLabs (free), Gamma; plus built-in screen readers and caption tools.

Final submission

- Research report
- Inclusive-content checklist
- Tool test log
- Accuracy findings
- Practical guide
- Quiz and answer key
- Presentation
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 79 — Capstone Showcase: Portfolio Website and Demo Day Kit

Project Question

How can AI help present a student's best projects in a professional portfolio and a confident demo?

What you have to do

1. Choose the target audience (final-year CSE students).
2. Research portfolio best practice and how to present technical projects to recruiters using at least two AI research tools.
3. Collect and verify reliable sources (recruiter guides, GitHub Pages docs, public developer portfolios (for style analysis only)).
4. Create a research summary and a portfolio quality checklist.
5. Select 4 best projects and write case-study summaries (problem, approach, result).
6. Build a portfolio on GitHub Pages or Notion.
7. Write a 3-minute demo script and 10 likely Q&A answers.
8. Create a 5-slide demo deck and a short recorded demo.
9. Automate one step (e.g. a workflow that updates a project list from a spreadsheet).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: describe only real work, give credit to teammates, protect private or confidential code.

Required AI skills

At least 6 from Topics 1–9. Suggested: Prompting + writing + web publishing + verification + presentation + video + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Notion, GitHub Pages, Gamma, HeyGen (free), ElevenLabs (free), Make.

Final submission

- Research report
- Portfolio quality checklist
- Case studies
- Portfolio site
- Demo script and Q&A
- Demo deck
- Recorded demo
- Automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks

CAPSTONE 80 — Fake Job Offer and Internship Scam Awareness Kit for CSE Freshers

Project Question

How can AI help CSE students quickly spot fake job offers, fake internships and fake recruiters, and verify an opportunity before sharing documents or money?

What you have to do

1. Choose the target audience (first-year and final-year CSE students looking for internships and jobs).
2. Research common fake job, fake internship and fake recruiter scam patterns (advance-fee, "registration fee", fake offer letters, fake interview links) using at least two AI research tools.
3. Collect and verify reliable sources (cybercrime.gov.in advisories, CERT-In, Ministry of Labour and state police public advisories, official company career-page guidance, LinkedIn and Naukri safety pages).
4. Create a research summary and a "verify before you apply" checklist.
5. Write 6 fictional scam messages (email, WhatsApp, Telegram) labelled "training samples", each with red flags marked and the correct safe response.
6. Write an 800-word guide on how to verify a recruiter, an offer letter, a company domain and a payment request.
7. Create a poster or infographic comparing a genuine offer process with a fake offer process.
8. Create a 5-slide campus talk and a 10-question quiz with answer key.
9. Automate one step (e.g. a workflow that takes a pasted job message and drafts a red-flag report from the checklist).
10. Test AI outputs for hallucinations, wrong or unsafe advice and outdated information.
11. Document all important prompts.
12. Explain ethics: use only fictional companies, recruiters and numbers; never expose real victims or real scammers' details; never ask AI to write a scam message that could be used for real; tell students to report fraud to the official portal and the placement cell.

Required AI skills

At least 6 from Topics 1–9. Suggested: Research + prompting + writing + verification + image generation + quiz design + presentation + automation + responsible AI.

Required tools

At least 4 free AI tools. Suggested: ChatGPT, Gemini, Claude, Perplexity, NotebookLM, Leonardo.ai, Gamma, Make/Zapier, Notion; plus Google Sheets for the red-flag scoring table.

Final submission

- Research report
- "Verify before you apply" checklist
- Scam message set with red flags
- Verification guide
- Comparison infographic
- Quiz and answer key
- Presentation
- Red-flag report automation
- Prompt library
- Verification report
- Before/after analysis
- Final demo

Weightage: 100 marks